

Zhouhao Zhang

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Education

Beihang University Sep 2020 – Jun 2024
Bachelor of Engineering in Automation (GPA: 3.80 / 4.00) Beijing

Internship

King Abdullah University of Science and Technology (KAUST) Apr 2025 – Sep 2025
Research Intern, Supervised by Prof. Shinkyu Park Jeddah, Saudi Arabia

- Developed ray tracing algorithm for octree map coverage computing.
- ROS2 workspace of **Crazyflie** for racing drone RL and multi agent RL.

Cytoderm Dec 2024 – Mar 2025
Robotics Engineer Beijing

- Developed unmanned intelligent production lines with JAKA AGV and Rokae robotic arms.

Zhipu AI X-lab Jun 2024 – Dec 2024
Robotics Research Intern, Supervised by Dr. Tong Yang Beijing

- Developed the **Scene Iconicity Graph** for enhancing robotic operations, utilizing foundation models' zero-shot capabilities to improve scene understanding and reasoning
- Integrated and deployed a **3D LiDAR SLAM** system and **vision-language navigation** on the Unitree Go2 quadruped robot, utilizing **behavior trees** to manage navigation tasks
- Leading the design of a **mobile research robot** equipped with multiple sensors and AI computing resources for advanced **embodied-AI experimentation**

Skyforce Technology Jul 2023 – Dec 2023
R&D Engineer Intern Shenzhen

- Pending **National Invention Patent**: Automated keystone correction for projectors with structured light technology
- Calibrated structured light systems through **Gray code** and **local homography** for enhanced accuracy
- Developed an automatic keystone correction algorithm incorporating **TOF** and **IMU**
- Implemented obstacle avoidance for projectors using **LiDAR** and the **RANSAC algorithm**

Skills

Languages: Chinese (Native), English (TOEFL: 98)
Programming Languages: Python, C, C++
Robot Development Environments: Linux, ROS, Gazebo, PyBullet, Isaac Sim, Mujoco
Development Libraries: Pytorch, OpenCV, OpenAI Gym, PCL, Eigen
Robot Hardware: UAV, Wheeled/Quadruped Chassis, Robotic Arms, LiDAR, Depth Camera, IMU, TOF
Embedded Systems: NVIDIA Jetson, Raspberry Pi, STM32, ESP32

Honors

National Scholarship | *National 0.2%, The Ministry of Education of the People's Republic of China* Dec 2022
First Prize, 23rd China University Robot Competition (ROBOCON) | *National Third Place* Jul 2024
Second Prize, 23rd China University Robot Competition (RoboMaster) Apr 2024
Second Prize, 22nd China University Robot Competition (ROBOCON) Jul 2023
First Prize, 2022 Five Provinces of North China University Robot Competition Nov 2022
First Prize, 2022 Beijing University Robot Competition Nov 2022
Second Prize, China Intelligent Robot Fighting and Gaming Competition Mar 2023
Third Prize, 38th National College Physics Competition Dec 2021
Outstanding Graduate Award | *Beihang University* Jun 2024
CATIC Scholarship | *Total 10 Places (0.3%) in Beihang University* Apr 2024
First Prize, Scholarship in Discipline Competition | *Beihang University* Apr 2024

Second Prize, Scholarship in Outstanding Social Work <i>Beihang University</i>	Apr 2024
Outstanding Student Leader <i>Beihang University</i>	Nov 2023
Top Prize, Learning Excellence Scholarship <i>Top 2%, Beihang University</i>	Dec 2022
Outstanding Student <i>Top 1%, Beihang University</i>	Sep 2022
Third Prize, 32nd FengRu Cup Competition & Yuyuan Robots Competition <i>Beihang University</i>	Jun 2022

Projects

MedAIR, Chinese University of Hong Kong <i>Graduation Project (Remote), Guided by Prof. Qi Dou</i> - Task planning of 7DOF da Vinci Surgical Robot via demonstration-guided reinforcement learning and policy chain - Surgical robot visual manipulation policy learning using world model-based reinforcement learning - High-level task planning of robot based on large language models and behavior tree	Dec 2023 – Jun 2024
Biomechanics and Soft Robotics Lab, Beihang University <i>Research Member, Guided by Dr. Lei Li</i> Publication: An Aerial–Aquatic Hitchhiking Robot with Remora-Inspired Tactile Sensors and Thrust Vectoring Units, in <i>Advanced Intelligent Systems</i>, DOI: 10.1002/aisy.202300381 - Debugged flight control programs and deployed SLAM algorithms for aerial-aquatic UAV	Apr 2023 – Oct 2023
Pattern Recognition & Image Processing Course Project, Beihang University <i>Project Leader</i> - Segmentation of medical images (liver and fundus vessels) using U-Net - Image stitching using Scale-Invariant Feature Transform (SIFT) for feature matching - Grasp position detection for small electronic components based on vision transformer - Hand written characters classification based on convolutional neural network - Mixture Gaussian model of population constructed by Expectation-Maximization algorithm - Point cloud classification and part segmentation using PointNet, PointNet++ EEG-net based classification of motor imagery - Solving the eight-digit problem with graph search algorithms	Dec 2023
BR Robotics Team, Beihang University <i>Member & Coach</i> Visual auto-shoot algorithm based on YOLO and localization prior for racing robot in ROBOCON 2023 Decision making algorithm leveraged minimax search and finite state machine for racing robot in ROBOCON 2024 - Deployment of SLAM algorithms including Vins-Mono, Vins-Fusion, RTABMAP, ORB-SLAM, and FAST-LIO2 Camera pose estimation using the PNP algorithm and morphological operation Robot path planning using the A-star algorithm and grid map Technical guidance for the RoboMaster, ROBOCON and RoboCup competitions - Founded Beihang University's first RoboMaster team	Aug 2022 – Jun 2024

Extracurricular Experience

Red Bird Challenge Camp <i>Hong Kong University of Science and Technology (Guangzhou)</i> - Designed and built a factory simulation environment in Gazebo to simulate intelligent mobile detection robots - Implemented 2D SLAM for mobile robots using the Cartographer algorithm - Developed autonomous navigation and dynamic obstacle avoidance systems using the ROS Navigation stack	Jun 2024 Guangzhou
Artificial Intelligence & Machine Learning Program <i>National University of Singapore</i> - Developed an innovative Seq2Seq model using LSTM for accurate population forecasting - Achieved winning team in the final presentation, earning commendation from Prof. Mehul Motani	Aug 2023 Singapore